

Electronics build guide

Matek F405-WING V2 / ArduPlane / FlySky / five servos with nose steering

For the **Electronics Team**. Follow the steps in order and record actual readings in the build record.

Remove the propeller for wiring, firmware, calibration, servo and failsafe work. Refit it only for a separate controlled propulsion test after bench checks pass.

Equipment	Build assumption
Flight controller	Matek F405-WING V2; not V1, WTE or TE
Navigation	Matek M9 GPS; exact SKU and compass presence unconfirmed
Controls	Two aileron + two V-tail + one nose-steering servo for supplied gear; 9 g metal gear, model unknown
Power	GenX 4S 5200 mAh LiPo; ReadytoSky 40 A 2-4S ESC advertised as Opto with 5 V / 3 A BEC
Propulsion	Emax ECO II 2807 1300 KV + Gemfan 9045 CL three-blade; combination unqualified
Radio / airspeed	FlySky FS-i6X + FS-IA10B; Matek airspeed SKU unconfirmed

Work in this order

Identify parts -> build and meter harnesses -> configure firmware -> connect radio and sensors -> centre and test servos -> test failsafes -> install and balance -> qualify propulsion. Save a parameter file after each successful stage.

Five servos: two ailerons, two tail controls and one nose steering. Establish control throws, flight CG, speeds, gains and return altitude from the completed aircraft and measured tests.

Resolve these labels first

Connector shape and wire colour are not a pinout

Unknown	Safe branch before connection
M9 GPS SKU	Read PCB and module labels and obtain that Matek manual. UART module: TX to FC RX3, RX to TX3; match specified supply and ground. Add SDA/SCL only if an I2C compass exists. CAN-only module: stop; this FC has no native CAN. Never infer pin order from another M9.
Airspeed SKU	ASPD-4525: I2C, ARSPD_TYPE=1. ASPD-DLVR: labelled I2C connector, ARSPD_TYPE=9; its CAN connector is different. Verified I2C branch: DA2/SDA + CL2/SCL, ARSPD_BUS=1. Other SKU: use its manual and leave disconnected until identified. [1,6,7]
ESC signal plug	Determine whether red is BEC output, absent, or required optocoupler supply. If BEC output, depin and insulate individually; retain signal and signal ground. If input-side power is required, follow the exact manual. Do not guess from Opto in a listing.
9 g servo model	Confirm voltage, pulse range, current and torque. Use 5 V only if supported. Do not select 6/7.2 V from size or metal gears alone. Five servos cover the supplied controls and steering; quantity is not torque approval.
Battery / connectors	Record mass, dimensions, connector polarity and charge limits. Meter every adapter. Never connect the 4S main lead to receiver, GPS, servo rail or 5 V pad.

Make a connector record

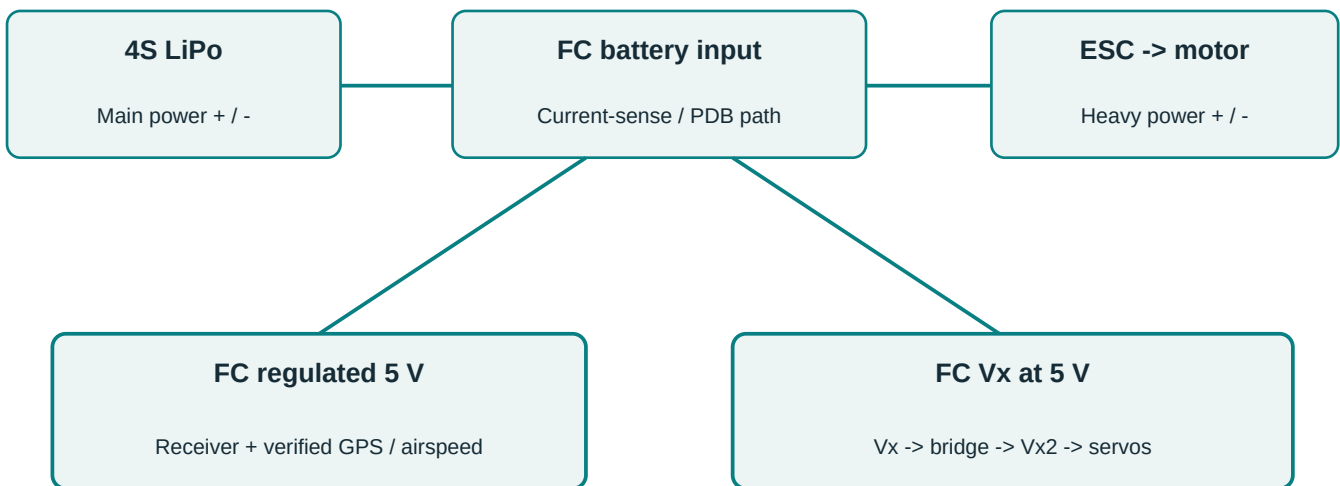
Draw each plug from a stated face. Number its cavities and label signals. With power removed, continuity-check both ends and label the harness.

Pad names below identify functions. Verify connector cavity numbers from the exact hardware manual before soldering.

Power architecture

One positive source per rail; common signal ground

Route battery and ESC power through the FC power-distribution pads. Match the V2 board layout [1].



ESC signal + ground -> FC S1 + G

ESC BEC output (if present) -> insulated; never joined to FC rails

Build and meter

1. Fit strain-relieved battery and ESC power leads to the documented pads. Keep high-current loops short. Select connectors and wire for measured load; insulate exposed joints.
2. Keep electronics 5 V and servo Vx positives separate. S5-S9 use Vx2: bridge Vx to Vx2 for this internal-BEC plan. Check before plugging in servos. [1]
3. If tests require an external BEC: remove the bridge, verify isolation, power Vx2 from that BEC and move all servo positives to that rail. Retain grounds; never join two BEC positives.
4. With peripherals unplugged, check polarity and hard shorts. First power through a suitable current limiter, then meter 5 V and Vx/Vx2 at far connectors. A limiter may brown out under servo load; it does not qualify flight power.

Wire the signal harness

FC outputs and transmitter channels are independent

FC connection	Destination / setting
S1 + G	ESC PWM signal + ground. SERVO1_FUNCTION=70 (Throttle). Isolate BEC output lead if present.
S2	Unused, SERVO2_FUNCTION=0; leave empty.
S3 + Vx + G	Left aileron. SERVO3_FUNCTION=4 (Aileron).
S4 + Vx + G	Right aileron. SERVO4_FUNCTION=4 (Aileron).
S5 + Vx2 + G	Left tail. Initially SERVO5_FUNCTION=79 (V-tail left). Validate page 8; physical motion governs.
S6 + Vx2 + G	Right tail. Initially SERVO6_FUNCTION=80 (V-tail right). Validate page 8.
S7 + Vx2 + G	Nose steering A39: SERVO7_FUNCTION=26 (GroundSteering). Omit only with a deliberately fixed nosewheel.
S8-S9 / LED	Unused; disconnect. Check SERVO8-10_FUNCTION for unwanted assignments.
RX2 + 5 V + G	FS-iA10B iBUS servo-output signal + supply + ground. Not SENS telemetry port. BRD_ALT_CONFIG=0.
RX3 / TX3 / supply / G	Verified UART GPS: module TX -> RX3, RX -> TX3; supply per exact SKU. SERIAL3_PROTOCOL=5 (GPS).
DA2 / CL2 / supply / G	Verified I2C compass / airspeed: SDA -> DA2, SCL -> CL2. Match pinout and supply. Check shared-bus address conflicts.

Custom output map: write and read back every assignment before connecting the ESC. S1 is throttle only after configuration. [2-4]

Start with supported 50 Hz PWM. Leave DShot disabled for this unidentified ESC. Check shared timer groups before changing rates. [2]

Firmware and first connection

Mission Planner on Windows; menu names can change

Install the matching target

1. Confirm **F405-WING V2**. Connect a data USB cable with battery, ESC and servos disconnected.
2. Install stable **ArduPlane / MatekF405-Wing**, version 4.4 or newer. Do not select Copter, INAV, F405-STD or F405-TE. [1]
3. With an ArduPilot bootloader, load the matching .apj. Otherwise follow the board DFU first-install procedure using firmware with its bootloader. [2]
4. Record target and version. Save initial parameters and inspect sensor messages. Insert a supported logging microSD with power off.

Configure before powering outputs

Configure: write page 4 settings in Full Parameter Tree/List. Reboot and read back. Use Mandatory Hardware for sensors, radio and modes; Servo Output for servos.

Do not bulk-import an internet parameter file. Keep arming checks enabled. Resolve each failed check instead of setting `ARMING_CHECK=0`. Save the parameters actually read from this board.

Recovery route

Recovery: disconnect propulsion power. Reconnect USB and restore the last known parameter file. Repeat sensor, output and failsafe checks after firmware changes.

Nose steering: centre A39 with `SERVO7_TRIM`; set safe endpoints and reversal so right rudder steers right. Push-test with motor off. Set `GROUND_STEER_ALT` deliberately. Stabilized steering needs healthy heading and `STEER2SRV_P` based on measured turning diameter. [14]

Radio input and modes

Plain controls from FS-i6X; mixing inside ArduPlane

Create an airplane model; select AFHDS 2A and bind the receiver. Disable transmitter V-tail/elevon mixes and centre trims. Use the receiver iBUS servo port with regulated 5 V. [5]

Input	Intended function
CH1	Roll; RCMAP_ROLL=1
CH2	Pitch; RCMAP_PITCH=2
CH3	Throttle; RCMAP_THROTTLE=3
CH4	Yaw; RCMAP_YAW=4
CH5	Three-position mode switch; FLTMODE_CH=5. Measure its actual detent values.
CH6 optional	Dedicated arm/disarm only after verifying the installed firmware option and behaviour; never share with mode control.

Calibrate

Calibrate inputs: power RX from verified FC 5 V. Check each stick moves the intended bar. Sweep sticks and switches; finish at low throttle with other sticks centred. Save. [8]

Input directions: right roll, pitch up, more throttle and right yaw must increase their normalized inputs. Correct RCn_REVERSED first; adjust surface direction later with SERVOx_REVERSED. Recalibrate after endpoint changes. [8]

Modes: initially set all six slots MANUAL. Assign the middle detent to FBWA. Keep the third MANUAL until navigation checks pass, then deliberately assign RTL. Check HUD mode at every detent and transition.

Receiver telemetry is not automatically flight-battery voltage. Keep antenna active ends clear of carbon and power wiring; range-test after final installation.

Sensors and battery readings

Calibrate in final orientation; inspect the actual response

Attitude, heading and position

Attitude: secure FC and set AHRS_ORIENTATION. Calibrate accelerometers in six positions, then level against the aircraft datum. Verify HUD roll/pitch follow the aircraft.

Compass: the FC has no onboard compass. Identify any external one, set orientation and calibrate away from metal and power wires. Verify heading in several known directions after installation. [2]

GPS: use SERIAL3_PROTOCOL=5 and the appropriate GPS type. Check outdoors for plausible position, fix and time. GPS-only hardware does not provide magnetic heading.

Battery monitor - V2 values

Parameter	Initial value
BATT_MONITOR / voltage / current pins	4 / BATT_VOLT_PIN=10 / BATT_CURR_PIN=11
BATT_VOLT_MULT / BATT_AMP_PERVLT	11.0 / 66.7; older-board 31.7 is wrong for V2
BATT_CAPACITY	5200 mAh nominal; validate usable capacity separately

Calibrate against a meter:

New voltage multiplier = old multiplier x meter voltage / reported voltage.

New current scale = old scale x reference current / reported current.

Use a steady meaningful current, not near zero. Check idle offset and consumed capacity. [1,4]

Set low/critical thresholds, delays, reserves and actions from pack specifications and measured loaded behaviour. Record them; the C-rating does not prove endurance.

Servos and V-tail direction

Left/right viewed from behind the aircraft, facing forward

Centre servos electrically before fitting horns. Align linkages mechanically; use SERVOx_TRIM for small corrections. Set conservative endpoints and check every hinge through full travel.

Test	Required response
MANUAL: roll right	Right aileron trailing edge up, left down; reverse for roll left.
MANUAL: pitch up	Both tail trailing edges move in upward pitch-producing sense; reverse for pitch down.
MANUAL: yaw right	Both tail trailing edges move toward aircraft right; reverse for yaw left.
FBWA, centred sticks: tilt aircraft right	Left aileron up, right down: correction opposes tilt.
FBWA: raise nose	Tail command produces nose-down correction; lowering nose produces nose-up correction.

V-tail correction: wrong in both pitch and yaw → reverse output. Wrong in only one axis → exchange functions 79/80 and repeat both tests. Physical motion governs the final mapping. [9]

Roll-to-rudder testing requires nonzero KFF_RDDRMIX; restore its intended value afterward. A stationary yaw test does not prove heading hold. [9]

Combined commands and supply loading

Test all four full pitch/yaw combinations. Check binding, buckling, buzzing and saturation. Adjust mechanics, endpoints and MIXING_GAIN; repeat MANUAL and FBWA tests. [9]

Load-test all servos together without prolonged stall. Capture voltage dips, current peaks, temperature and resets. A multimeter may miss brief dips. Limits: Vx 5 A continuous / 6 A peak; electronics 5 V rail 2 A. Verify margin. [1]

Airspeed and propulsion

Sensor response is not calibration; motor spin is not thrust

Airspeed branch

Identify sensor and set ARSPD_TYPE / ARSPD_BUS. ASPD-4525: type 1. DLVR-L10D I2C: type 9. Never connect its CAN socket to I2C. Reboot and check health. [6,7]

Put pitot in clean airflow, clear of prop wash. Match dynamic/static ports; avoid leaks and kinks. Let temperature settle. Shield from wind without sealing ports; run Preflight Calibration for zero. [10]

Apply gentle airflow: reading should rise and return near zero. Avoid pressure, liquid or blowing into tubing. Start ARSPD_USE=0; some Plane versions still use a healthy sensor for speed scaling. [10,11]

If readings are suspect, set ARSPD_TYPE=0 and resolve them. Ratio calibration needs a safely flying aircraft and log review. Bench airflow does not establish true airspeed or stall speed.

ESC and motor

Motor mount: four M3 holes on 19 mm PCD; adjacent spacing 13.435 mm. Check actual motor fit and safe screw engagement. Emax lists 6-7-inch props; test the proposed 9-inch combination before use. [15]

Confirm ESC PWM range and calibration method. FC/RX must stay powered while ESC power switches separately. Use a documented isolated setup or servo tester; do not backfeed power rails. [12]

If the ESC manual specifies high/low calibration: remove prop, switch ESC off, command maximum, power ESC, wait for its indication, then command minimum. Verify acceptance. Do not use multicopter calibration for Plane. [12]

Check stopped motor when disarmed and at minimum. Restrain the airframe for a brief low-speed rotation check. Power off before swapping two phase wires to reverse. Confirm pusher prop orientation produces forward thrust.

Failsafes: prove the trigger

Prop removed; disconnect motor power for configuration tests

With prop removed and telemetry recording, switch the transmitter off. Observe RC inputs, failsafe message, mode and recovery when the link returns.

Radio loss

Set `THR_FAILSAFE=1`. Check whether RX stops frames, flags loss or holds/presets channels. Held normal channels may hide link loss. Configure a detectable condition and repeat. [13]

For low-throttle detection, put `THR_FS_VALUE` between measured loss and valid-low throttle. Example only: loss 900, valid low 1000, threshold 950 us. Require clear separation.

Candidate policy: short loss `CIRCLE`, `FS_SHORT_ACTN=1`; sustained loss `RTL`, `FS_LONG_ACTN=1`. Verify firmware labels and both timeouts. Set valid home, return height, terrain clearance and power reserve before flight. [13]

Battery and navigation failures

Record low/critical voltage, capacity, delay and actions. Test with a suitable variable supply or temporary thresholds, then restore and read back flight values. Never deeply discharge the LiPo for a test.

A ground `RTL` mode change does not prove a safe return. Test navigation-failure policy in simulation, then controlled evaluation with an experienced pilot. Do not unplug live sensors near a running prop.

Record trigger, delay, action and recovery. Repeat from every intended flight mode. If a failsafe cannot be observed or explained, keep the flight gate closed.

Installation and release record

Complete on the actual aircraft

Restrain the adjustable battery. Separate power wiring from GPS, compass and receiver. Protect wires at edges, strain-relieve plugs and label wing/tail connectors. Keep USB, SD, rail bridge and battery access open.

Provisional setup balance station

Provisional balance mark: 522 mm aft of nose, or 22.2 mm aft of front joiner. This is geometric quarter-MAC, not an approved flight CG range. Check complete mass, control authority and final CG with an experienced fixed-wing pilot.

xCG = sum(mass x position) / total mass

Battery adjustment: battery shift = total mass / battery mass x desired CG shift.

Use one nose datum, aft positive. Include wiring, glue and finish. Mark repeatable pack positions; verify assembled balance and lateral balance.

Record	Measured / approved value
Firmware / saved parameters	_____
5 V minimum / Vx minimum / servo peak current	_____
Voltage calibration / current reference	_____
RC loss trigger / short / long / recovery	_____
Battery low / critical / actions	_____
Total mass / battery mass / actual CG from nose	_____
Desired CG shift / required battery movement	_____
Control throws / propulsion test / flight CG review	_____

Before flight: verify connectors, loaded supplies, control directions, range/failsafes, sensors, structure, CG and propulsion. Measure current, thrust, voltage and temperature on a proper restrained test fixture.

Manuals and reference links

Primary sources checked 19 September 2026

[1] Matek V2 specifications, layout and mapping

<https://www.mateksys.com/?portfolio=f405-wing-v2>

[2] ArduPilot F405-Wing (older details need V2 cross-check)

<https://ardupilot.org/plane/docs/common-matekf405-wing.html>

[3] ArduPilot Choosing Servo Functions

<https://ardupilot.org/plane/docs/servo-functions.html>

[4] ArduPilot board hardware definition (master)

https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-Wing/hwdef.dat

[5] FlySky FS-iA10B specification

<https://www.flysky-cn.com/ia10b-canshu>

[6] Matek ASPD-4525

<https://www.mateksys.com/?portfolio=aspd-4525>

[7] Matek ASPD-DLVR

<https://www.mateksys.com/?portfolio=aspd-dlvr>

[8] ArduPilot RC input mapping and reversal

<https://ardupilot.org/plane/docs/rc-throw-trim.html>

[9] ArduPilot V-tail / inverted V-tail setup

<https://ardupilot.org/plane/docs/guide-vtail-plane.html>

[10] ArduPilot Airspeed Calibration

<https://ardupilot.org/plane/docs/calibrating-an-airspeed-sensor.html>

[11] ArduPilot Airspeed Parameters Setup

<https://ardupilot.org/plane/docs/airspeed-parameters-setup.html>

[12] ArduPilot ESC Calibration

<https://ardupilot.org/plane/docs/common-esc-calibration.html>

[13] ArduPilot Plane Failsafe Function

<https://ardupilot.org/plane/docs/apms-failsafe-function.html>

[14] ArduPilot Ground Steering

<https://ardupilot.org/plane/docs/tuning-ground-steering-for-a-plane.html>

[15] Emax ECO II 2807 official drawing in product description

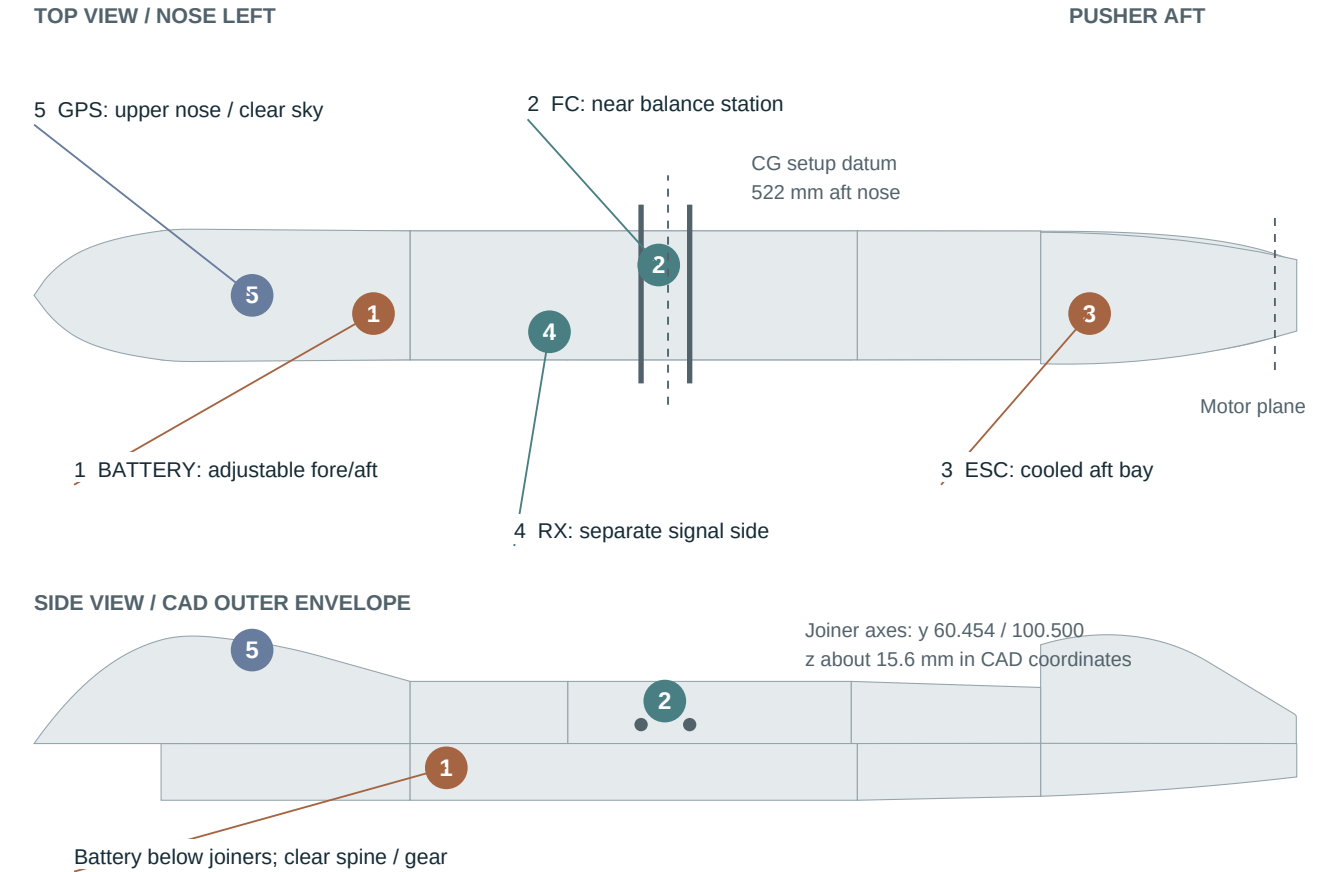
<https://emaxmodel.com/products/emax-eco-ii-series-2807-1300kv-1700kv-1500kv-brushless-motor-for-rc-drone-fpv-racing>

Keep the exact ESC, servo, GPS and airspeed manuals with the build record. Complete the measured bench and flight checks on the actual aircraft.

Where the electronics go

Proposed zones on the actual CAD body - not mounting-hole templates

With the structure open, trial-fit the battery and equipment in these zones. Check actual battery size and cable exit before making its tray or hatch.

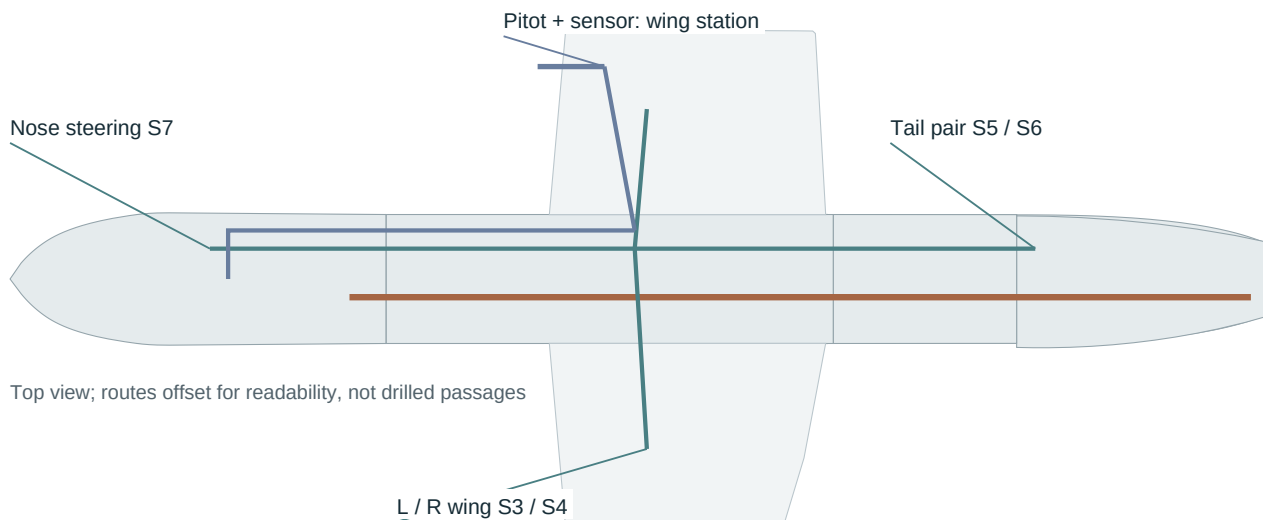


Zone	Mounting decision
1 Battery	Adjustable tray below the wing tubes, to one side of the centre wood spine as space permits. Clear nose/main gear and tube roof. Positive restraint in every direction; accessible connector. No pack dimensions or fitted tray claimed.
2 Flight controller	Accessible near the setup balance station on a firm mounting platform, using documented board grommets. Arrow forward; clear USB/SD access, level datum and wires that pull on the board.
3 / 4 / 5 ESC, RX, GPS	ESC aft with airflow and service access; RX on the signal side. Put M9 antenna upward with clear sky; if compass equipped, separate it from battery/current wiring and steel. Place RX active antenna ends clear of carbon and motor power. [16]

Wing pitot placement and harness routes are shown next. Colour marks identify functions; their drawn size and coordinates are not component footprints.

Harness routes and mounting

Keep removable joints accessible; confirm fit before closing skins



Harness	Route and service point
Brown: high current	Battery -> FC distribution -> ESC -> motor. Keep + and - together and away from signal/sensor bundles. Keep ESC input leads short; obtain the actual ESC maker cable-length/capacitor guidance before extending. No unsupported length limit is assumed.
Teal: control	Separate 3-wire S3/S4 wing leads, labelled L / R with signal, servo positive and ground. Retain keyed connectors at wing roots. Route S5/S6 to the tail and S7 to nose steering; secure leads clear of hinges, pushrods and gear.
Blue: navigation	GPS UART and optional compass to FC; airspeed I2C to a supported sensor near its pitot. Mount probe upstream of the wing leading edge in undisturbed flow, away from fuselage boundary layer and prop wash. Keep tubes short, unkinked and removable. Do not choose a fuselage-roof position just because it is convenient. [17]

Install in six passes

1. Trial-fit equipment and mark hatch access.
2. Check battery travel around spine, gear and joiners.
3. Install mounts; separate power and signal wiring.
4. Fit labelled service plugs and strain relief.
5. Check restraint, cooling, linkage travel and antennas with hatches shut.
6. Repeat loaded-power, range, sensor, direction and balance checks.

Trial-fit actual components. The shaded zones are suggested locations, not mounting-hole templates or guaranteed clearances.

[16] ArduPilot GPS/compass mounting

[17] ArduPilot Pitot Tube Considerations